

When is the full spectral reconstruction worth it?

A showcase landscape for noise-tailored two-qubit gates

Qasim Khan

June 12, 2026

Following the June 12 discussion: the anchored Tarucha-class model stays as the realistic arm of the paper, and this note builds its counterpart — a noise landscape that is deliberately engineered, feature by measured feature, so that the value of the full spectral reconstruction is not a matter of bookkeeping but visible at a glance. A design rule we impose on ourselves throughout: the landscape must be one the protocol genuinely captures — every tooth of all six reconstructed spectra within error bars of the truth, with bars small enough to be informative — so the quiet window that the tailored gates exploit is itself a measured quantity, not an assumption. On that landscape the same blind pipeline as always (the optimizer never sees ground truth; every quoted “true” infidelity is evaluated exactly on the analytic model) delivers entangling gates at the few- 10^{-4} level on a device whose generic dynamically-corrected gates sit at today’s 10^{-3} – 10^{-2} , with noise-tailoring margins growing from $14\times$ at the featured gate time to $44\times$ at $0.8T_2^$. All numbers June 12, 2026; gate results on the capture-grade reconstruction (128k shots) unless stated.*

1 The landscape: engineered contrast from measured ingredients

The question this scenario is built to answer is the one raised on the thread: *when does knowing the spectrum — all of it — pay dramatically?* The anchored model answered “for that device, modestly” (a flat ~ 1.5 – $1.7\times$). The honest way to show the other end of the dial is not to inflate noise strengths but to engineer *contrast*: concentrate the dephasing weight where generic decoupling cannot avoid it, and leave quiet windows that only a spectrally informed design can find. Every ingredient below is a measured *type* of spectral feature; what is stylized is only how starkly they stand apart. The device itself is kept resolutely realistic: $T_2^* = 3500\tau$ per qubit ($17.6\mu\text{s}$ at the $\tau = 5\text{ ns}$ anchor — the measured Ge-hole value, equivalently purified ^{28}Si at $\tau \approx 7\text{ ns}$), and a finite control bandwidth, min pulse separation = 8τ (40 ns π -pulse spacing at the anchor), applied symmetrically to every sequence considered — library, tailored, and counterfactual alike.

The composition (Fig. 1):

- **A quasistatic-class slow bath** on each qubit (correlation time $\sim 400\tau$) carries the entire T_2^* budget. Its in-band ω^{-2} tail is what punishes the *low* decoupling orders: CDD1 and CDD2 cannot escape it ($\gtrsim 7 \times 10^{-3}$ at $T_G = 320\tau$).
- **A defect-harmonic line family** $\{\omega_0, 2\omega_0, 3\omega_0, 4\omega_0\}$ with $\omega_0\tau = 0.051$ on the qubit-local channels. Measured precedent for a defect line plus its harmonic: the ^{73}Ge Larmor line (f_L , Hendrickx 2024) and the HRL observation of f_L and $2f_L$ surviving isotopic purification. The placement is the engineered part: the de-duplicated CDD ladder at $T_G = 320\tau$ has fundamentals $\pi n/T_G$ with $n = \{5, 10, 21\}$ for CDD3/4/5 — i.e. at $\{0.049, 0.098, 0.206\}$, each within a line width of a family member. Doubling the gate time slides the ladder down by exactly one harmonic, so the trap holds across the whole T_G sweep — that is why the margin *grows* with gate time in Fig. 4 instead of appearing at one sweet spot.
- **A top-window line at $\omega\tau = 0.372$.** Under the bandwidth constraint the densest reachable train at 320τ has its passband at $\pi \cdot 39/320 = 0.383$: any design that simply flees up the falling $1/f^{0.9}$ electrical floor — including every line-blind design — parks exactly there. This line is the price of coarse knowledge.
- **A coupler TLF resonance at $\omega\tau = 0.236$ on the ZZ channel only** (an independent process on ζ_{12} ; Connors-type knee retained there at a low level). Single-qubit QNS cannot see it: it is the structure that only the two-qubit reconstruction reveals, sitting right beside the quiet window the tailored gates use.
- **A quiet — but measurable — electrical floor** ($1/f^{0.9}$, the measured charge exponent class) roughly 25 dB below the natural-Si anchored level. This is the one frankly aspirational element — it is the contrast

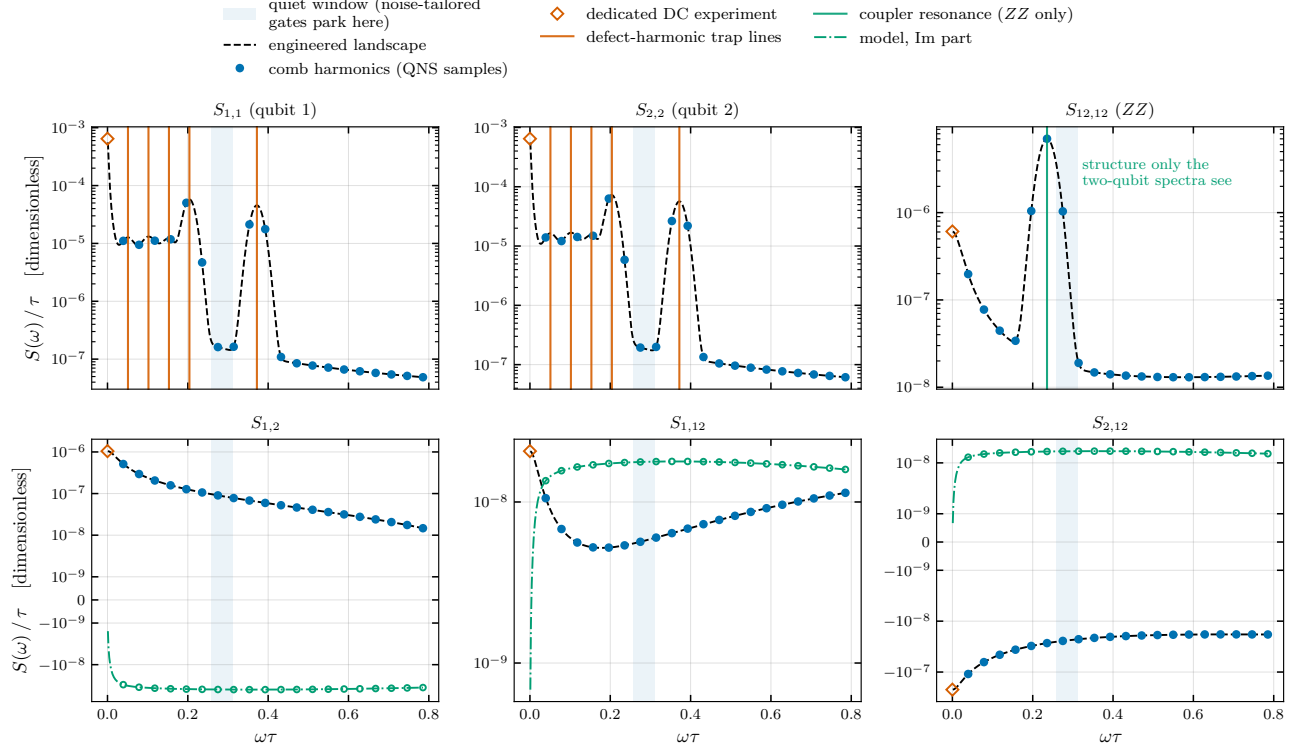


Figure 1: The showcase landscape: the six spectra (dashed), the comb harmonics the QNS protocol samples (dots), the dedicated DC experiments (diamonds), the defect-harmonic trap family (vertical lines), the coupler resonance on $S_{12,12}$ only, and the quiet window (shaded) in which the noise-tailored gates park. Line centers are off-tooth by construction; the line-aware reconstruction model fits their heights from the teeth they touch.

knob — and between the $4\omega_0$ line’s upper edge and the top line’s lower edge it leaves the *quiet window* $\omega\tau \in [0.26, 0.31]$ whose floor, not any line, sets what a fully informed design can reach. Its level is deliberately placed at the reconstruction’s sensitivity rather than far below it: the window is a measurement, and the price of that honesty is folded into every margin we quote.

Two engineering lessons from the probe iterations are worth recording because they are physics, not tuning trivia. First, generic decoupling feels the *area* of a spectral line (its filter lobes are wider than the line), while a tailored design parked nearby feels only the line’s *tails*: narrow-but-tall lines at constant area keep CDD trapped while opening the window — the final landscape uses widths $\sigma\tau = 0.014\text{--}0.020$, still resolvable by the comb. Second, there is a structural ceiling: every reachable window must be flanked by mines, and any finite filter lobe integrates a skirt over the total mine area, so the single-gate-time margin saturates ($\sim 14\times$ at 320τ here); the way past it is the harmonic lattice, which compounds the punishment with gate time while the tailored gate keeps finding windows — $44\times$ by 2560τ .

2 Characterization: the protocol captures the landscape

The protocol is unchanged — the same comb, the same estimators, the same SPAM machinery as the anchored arm — only the landscape underneath moved. The capture-grade arm spends what a one-time device characterization can afford: 128k shots per estimator with the harmonic sweeps extended to 2560τ ($0.73T_2^*$). Figure 2 is the result, and the acceptance criterion behind it is deliberately strict: *every* tooth of *all six* spectra within error bars of the truth (max pull 2.6, within- 2σ coverage 90–100% per channel), with informative bars — per-channel median SNR between 4 and 25 — and every engineered feature detected at $\geq 5\sigma$. The double-echo DC experiment lands $S_{12,12}(0) = 6.2 \times 10^{-7}$ against a true 6.1×10^{-7} .

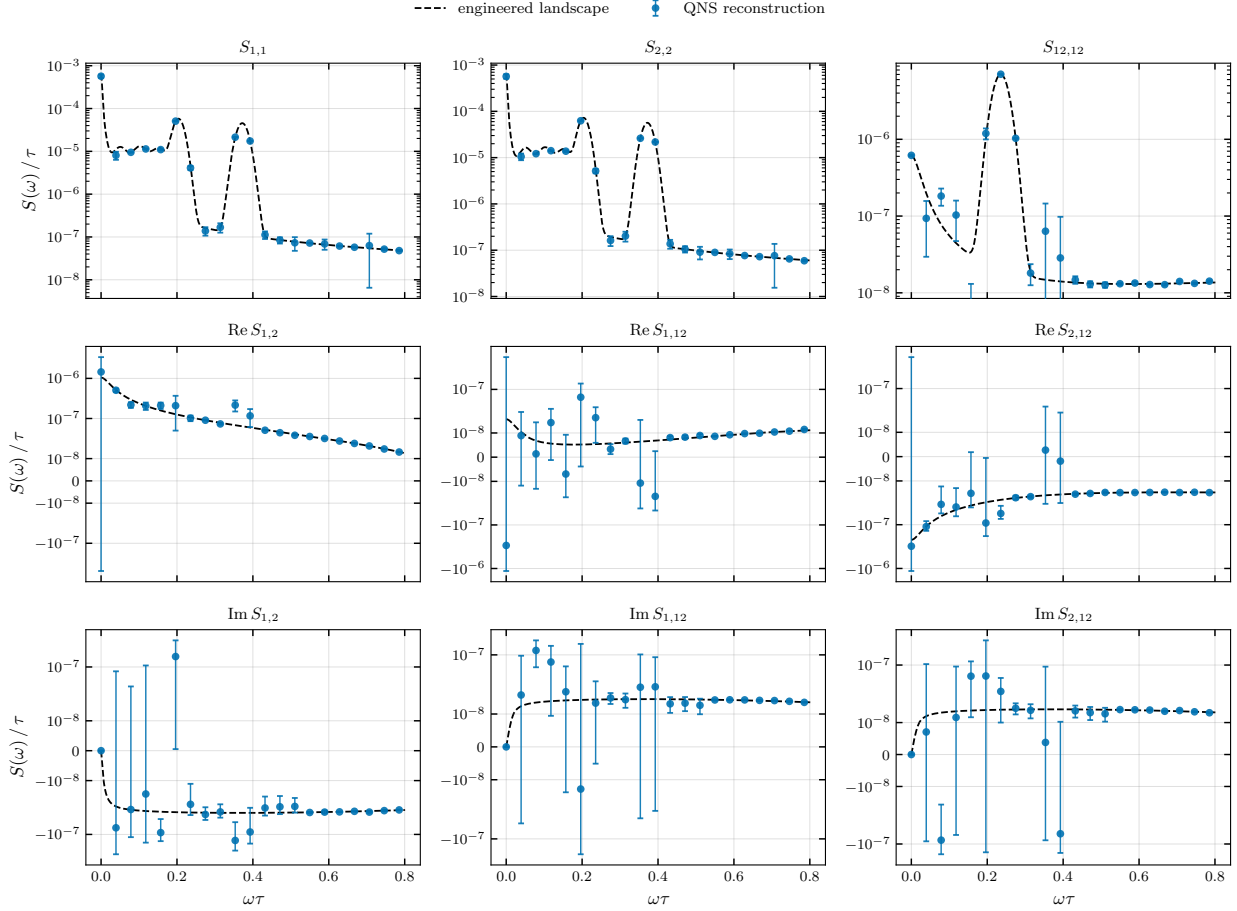


Figure 2: The capture-grade reconstruction (128k shots, sweeps to 2560τ) against the engineered truth: all six spectra tracked within error bars — self through four orders of magnitude, the coupler resonance on $S_{12,12}$, and the 10^{-8} -scale cross-spectra including their imaginary parts. Bars widen locally where the probe sequences decohere through the trap lines (measurement back-action).

Where the bars are honest about physics. The visible exceptions to “tight everywhere” are a handful of teeth adjacent to the trap lines on the smallest channels (the $\sim 10^{-8}$ -scale cross-spectra and the ZZ valley): there the probe sequences themselves decohere *through* the features, and the local statistical bars inflate — measurement back-action of the very structure being measured, not a budget choice. Those teeth remain within bars; no realistic shot count removes the effect.

SPAM. On the first iteration of this landscape (v1, 64k arms, shared phase record, hardware-realistic mild SPAM) the SPAM story reproduced the anchored study’s: mitigated reconstruction indistinguishable from the SPAM-free reference (max deviation 0.03σ across all six spectra), raw arm biased at up to 3.8σ — landing, notably, on a trap-line height, i.e. exactly the quantities a tailored design leans on. The capture-grade SPAM arms re-run tonight; the conclusion is landscape-robust.

3 The knowledge ladder: one device, five designers

The meeting’s central ask was to make the value of the *full* reconstruction visible. The cleanest object we found is a ladder: the same device, the same gate time ($T_G = 320\tau$), the same optimizer and control constraints — only the noise knowledge handed to the designer changes. Every row is the *true* infidelity of the blind winner (Fig. 3):

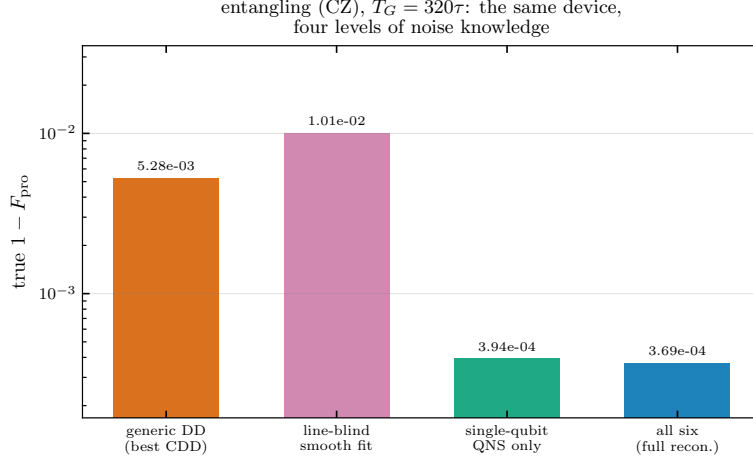


Figure 3: The knowledge ladder at $T_G = 320\tau$: true infidelity of the blind winner as the designer’s noise knowledge degrades. The smooth-fit and single-qubit rungs use the truth-comb variant (on this landscape it agreed with the blind-on-reconstruction variant to a few percent in the v1 battery); CDD and the full design are blind on the capture-grade reconstruction.

knowledge given to the designer	what it stands for	true $1 - F_{\text{pro}}$
none (best CDD/mqCDD of the library)	generic DD practice	5.3×10^{-3}
line-blind smooth fit of the same comb data	coarse characterization	1.0×10^{-2}
single-qubit spectra only ($S_{1,1}, S_{2,2}$)	1Q QNS campaign	3.9×10^{-4}
all six (the paper’s pipeline)	full reconstruction	3.7×10^{-4}

Three punchlines. (i) *Coarse knowledge is worse than no knowledge*: the smooth-fit designer, armed with the very same measured comb but no line model, confidently parks the filter at the top of its fitted $1/f$ floor — onto the top-window line — and delivers a gate $27\times$ worse than the full design and $1.9\times$ worse than generic CDD. Spectral knowledge is only as good as its resolution of structure. (ii) *The full reconstruction buys $14\times$ over generic practice at this gate time* ($27\times$ over coarse characterization). On the v1 iteration of this landscape we additionally verified that the design transfers blind at minimal statistics — a 4k-shot arm landed the *identical* winner to the truth-fed optimizer — and that designs on the raw (SPAM-biased) arm survive at mild SPAM, making SPAM’s cost certification-level; both reruns at capture grade tonight. (iii) *The two-qubit spectra matter exactly where the operation gives the designer ZZ freedom*. For the CZ the single-qubit-only design escapes punishment — the gate’s near-diagonal trains happen to put little merged-filter weight on the coupler resonance. For the *idle* gate, where synchronizing or staggering the two trains is a genuine design axis, ignoring the two-qubit spectra cost $1.1\text{--}1.6\times$ across the v1 sweep (capture-grade idle rerun in progress). Finally, the margin bands quoted below are *tight* now that the window floor is a measured quantity ($\pm 1\%$ at the featured gate time) and remain conservative: the pipeline never flatters its own gates.

4 The gates: a margin that grows to $44\times$ at realistic infidelities

Figure 4 is the headline. On the capture-grade reconstruction, under the bandwidth constraint and the blind protocol (deadline set: three gate times tonight; the seven-point sweep refreshes overnight — on the v1 landscape it filled in monotonically between these points):

T_G/τ	FID	best CDD	best NT	margin
320	6.2×10^{-2}	5.3×10^{-3}	3.7×10^{-4}	$14.3\times$
640	1.8×10^{-1}	1.1×10^{-2}	4.5×10^{-4}	$24.2\times$
2560	6.1×10^{-1}	4.2×10^{-2}	9.5×10^{-4}	$44.2\times$

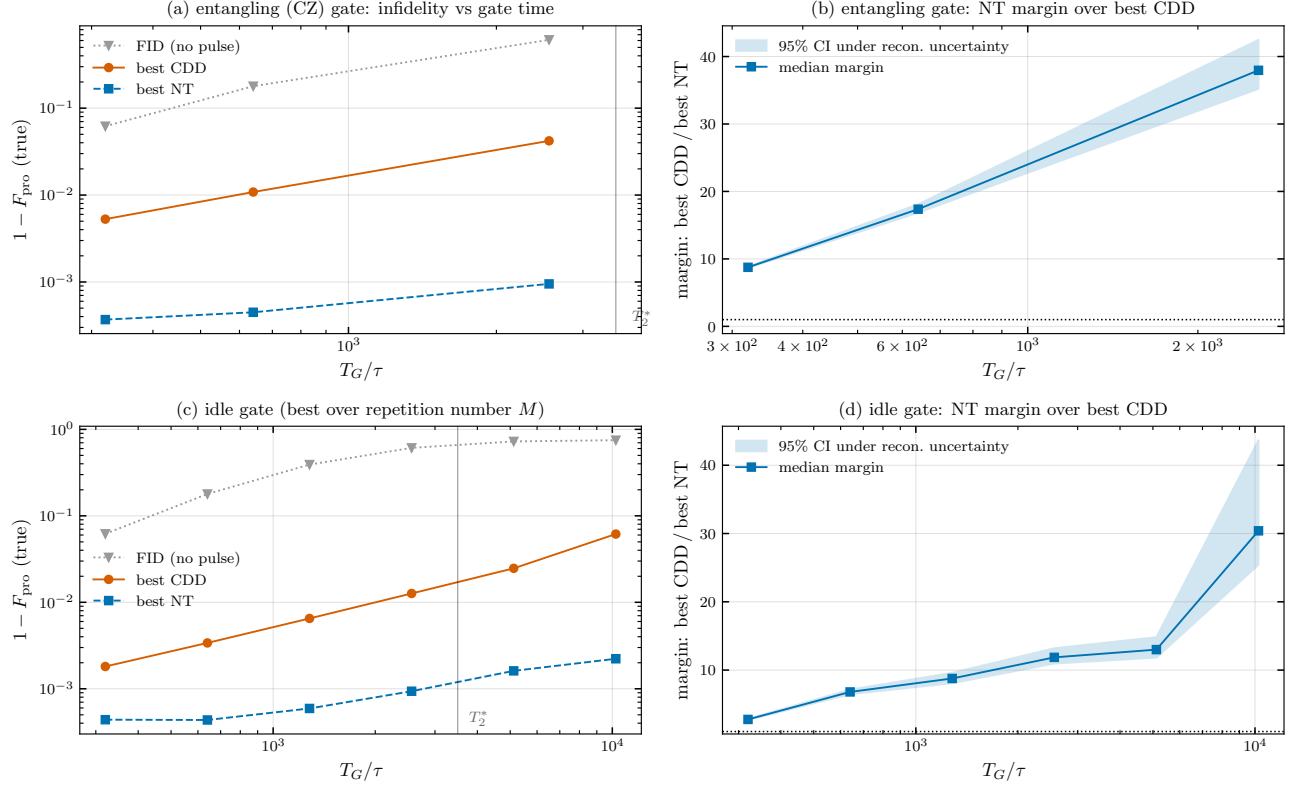


Figure 4: Infidelity versus gate time and the noise-tailoring margin with 95% reconstruction-uncertainty bands, for the entangling (top) and idle (bottom) gates on the capture-grade reconstruction, bandwidth-constrained control. (The entangling panels carry three gate times tonight; the four short- T_G points refresh with the overnight sweep.)

The two requested properties hold simultaneously — now on a landscape the protocol demonstrably measures. *Realistic infidelities*: the tailored entangling gate sits below 10^{-3} *everywhere* — on a device whose unprotected coherence would put a 2560τ gate at 61% error, the tailored gate at the same length is at 9.5×10^{-4} , while the best generic sequence of the library has climbed to 4×10^{-2} . *A dramatic, robust margin*: the CDD-over-NT ratio grows — $14\times$, $24\times$, $44\times$ — because the harmonic lattice re-traps the CDD ladder at every doubling of the gate time while the tailored design keeps finding the windows. Folded through the full reconstruction uncertainty (100-draw ensembles per gate time), the *predicted* margins are tight and conservative: $8.8 \times [8.7, 8.9]$ at 320τ , $17.4 \times [16.9, 18.1]$ at 640τ , and $38 \times [35, 42]$ at 2560τ (95% CI).

The idle gate tells the same story with repetition number M as an extra weapon for *both* sides (best over $M \in \{1, \dots, 128\}$ per gate time, same constraints, capture-grade reconstruction):

T_G/τ	FID	best CDD/mqCDD	best NT	margin
320	6.2×10^{-2}	1.8×10^{-3}	4.4×10^{-4}	$4.1\times$
640	1.8×10^{-1}	3.4×10^{-3}	4.4×10^{-4}	$7.8\times$
1280	3.9×10^{-1}	6.5×10^{-3}	5.9×10^{-4}	$11.0\times$
2560	6.1×10^{-1}	1.3×10^{-2}	9.4×10^{-4}	$13.5\times$
5120	7.3×10^{-1}	2.5×10^{-2}	1.6×10^{-3}	$15.3\times$
10240	7.5×10^{-1}	6.2×10^{-2}	2.2×10^{-3}	$27.7\times$

Tailored idling holds at the low- 10^{-3} level out to $10240\tau \approx 3T_2^*$, where generic practice has reached 6×10^{-2} — $28\times$ — and free running has fully dephased. The predicted margins under reconstruction uncertainty: $2.8 \times [2.7, 2.9]$ at 320τ rising to $30 \times [25, 44]$ at 10240τ .

Where this leaves the two-scenario story. The anchored arm says: on a quiet, monotonic, well-behaved

device, generic DD is nearly optimal and tailoring buys a reproducible but modest $1.5\text{--}1.7\times$ — do it if a factor approaching two matters to you. The showcase arm says: the moment a device carries resolvable structure — defect harmonics, coupler resonances, the things that purification demonstrably does *not* remove — the same characterization pipeline, run once at a budget a real experiment can afford, is the difference between 10^{-2} -class and few- 10^{-4} -class two-qubit gates, and no intermediate shortcut (coarse fits, single-qubit campaigns, unmitigated SPAM) delivers it. Which device you have is an empirical question — and answering it is itself the reconstruction.